

The Autonomic Nervous System

Human Physiology Chapter 9

Teaching Materials

December 24, 2025

Outline

9.1 Neural Control of Involuntary Effectors

9.2 Divisions of the Autonomic Nervous System

9.3 Functions of the Autonomic Nervous System

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Introduction to the Autonomic Nervous System

- The **autonomic nervous system** (ANS, 自主神经系统) regulates involuntary functions
- Controls **cardiac muscle** (心肌), **smooth muscles** (平滑肌), and **glands** (腺体)
- Differs from somatic motor system which controls voluntary skeletal muscle
- Essential for maintaining **homeostasis** (体内平衡)

Two-Neuron Pathway

- Autonomic motor control uses a unique **two-neuron pathway** (双神经元通路)
- **Preganglionic neuron** (节前神经元): Cell body in CNS, axon synapses in ganglion
- **Postganglionic neuron** (节后神经元): Cell body in ganglion, axon reaches effector
- Synapse occurs in an **autonomic ganglion** (自主神经节) outside the CNS



Comparison: Somatic vs Autonomic

- **Somatic motor system:**
 - Single neuron from CNS to skeletal muscle
 - No ganglia
 - Always excitatory using acetylcholine
 - Voluntary control
- **Autonomic motor system:**
 - Two neurons with ganglion synapse
 - Innervates cardiac/smooth muscle and glands
 - Can be excitatory or inhibitory
 - Involuntary control



Visceral Effector Organs

- **Visceral effectors** (内脏效应器) maintain function after nerve damage
- Smooth muscles have **resting tone** (静息张力) without nerve stimulation
- **Denervation hypersensitivity** (去神经超敏): Increased sensitivity after nerve damage
- **Intrinsic rhythmicity** (内在节律性): Cardiac and smooth muscles contract without nerves
- Autonomic innervation modulates (increases or decreases) this intrinsic activity

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Overview of Divisions

- Two main divisions: **Sympathetic** (交感) and **Parasympathetic** (副交感)
- Differ in:
 - Origin of preganglionic neurons
 - Location of ganglia
 - Target organs
 - General functions
- Often produce opposite effects on the same organ (**antagonistic**, 拮抗的)

Sympathetic Division: Thoracolumbar Origin

- Also called **thoracolumbar division** (胸腰分部)
- Preganglionic neurons originate from spinal levels **T1 to L2**
- Two types of ganglia:
 - **Paravertebral ganglia** (椎旁神经节): Sympathetic chain parallel to spinal cord
 - **Collateral ganglia** (副神经节): In abdominal cavity
- Mediates **fight-or-flight response** (战斗或逃跑反应)

Sympathetic Chain of Ganglia

- Paired chains of **paravertebral ganglia** (椎旁神经节) run parallel to spinal cord
- Connected to form **sympathetic chain** (交感链)
- Extends from cervical to sacral regions
- Preganglionic fibers can synapse at different levels
- Allows for widespread, coordinated responses



Sympathetic Pathways

- **White rami communicantes** (白交通支): Myelinated preganglionic fibers enter chain
- **Gray rami communicantes** (灰交通支): Unmyelinated postganglionic fibers exit chain
- Three pathway options:
 - Synapse at same level, return via gray ramus
 - Synapse at different level in chain
 - Pass through chain to collateral ganglion



Collateral Ganglia

- Located in abdominal cavity, unpaired
- Include:
 - **Celiac ganglion** (腹腔神经节)
 - **Superior mesenteric ganglion** (肠系膜上神经节)
 - **Inferior mesenteric ganglion** (肠系膜下神经节)
- Receive preganglionic fibers via **splanchnic nerves** (内脏神经)
- Innervate digestive organs and other abdominal viscera



Sympathetic Division Summary

- **Divergence** (发散): One preganglionic fiber contacts many postganglionic neurons
- **Convergence** (汇聚): One postganglionic neuron receives input from multiple preganglionic fibers
- Results in **mass activation** (大规模激活): Coordinated system-wide response
- Widespread effects on multiple organs simultaneously



The Adrenal Medulla

- **Adrenal medulla** (肾上腺髓质) is a modified sympathetic ganglion
- Preganglionic sympathetic fibers directly innervate medulla cells
- Secretes hormones into bloodstream:
 - **Epinephrine** (肾上腺素, adrenaline): 85%
 - **Norepinephrine** (去甲肾上腺素): 15%
- Provides prolonged, systemic sympathetic effects
- Key component of **sympathoadrenal system**

Parasympathetic Division: Craniosacral Origin

- Also called **craniosacral division** (颅骶分部)
- Preganglionic fibers originate from:
 - Cranial nerves: III, VII, IX, X
 - Sacral spinal cord: S2-S4
- Ganglia are **terminal ganglia** (终末神经节) near or in target organs
- Long preganglionic fibers, short postganglionic fibers
- Mediates **rest-and-digest** functions

Parasympathetic Division Summary

- Cranial nerves provide parasympathetic innervation to head and thoracoabdominal organs
- Sacral nerves innervate pelvic organs
- More localized, organ-specific responses than sympathetic
- Promotes energy conservation and restoration



Complete Autonomic System Overview

- **Sympathetic** (red): Thoracolumbar origin, ganglia near spinal cord
- **Parasympathetic** (blue): Craniosacral origin, ganglia near organs
- Solid lines: Preganglionic fibers
- Dashed lines: Postganglionic fibers
- Most organs receive dual innervation from both divisions



The Vagus Nerve

- **Vagus nerve** (迷走神经, cranial nerve X) is the major parasympathetic nerve
- Originates in medulla oblongata
- Provides extensive innervation:
 - Heart (slows heart rate)
 - Lungs (constricts bronchioles)
 - Digestive organs (promotes digestion)
 - Liver and pancreas
- Name means “wandering” due to its extensive distribution



Summary of Autonomic Effects

- Comprehensive overview of autonomic effects on various organs
- Shows sympathetic vs. parasympathetic effects
- Demonstrates antagonistic, complementary, and cooperative relationships
- Key reference for understanding dual innervation patterns
- Important for clinical applications and understanding drug effects



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Neurotransmitters of the ANS

- **Cholinergic neurons** (胆碱能神经元): Release **acetylcholine** (ACh, 乙酰胆碱)
 - All preganglionic neurons (both divisions)
 - All postganglionic parasympathetic neurons
 - Some postganglionic sympathetic neurons
- **Adrenergic neurons** (肾上腺素能神经元): Release **norepinephrine** (NE, 去甲肾上腺素)
 - Most postganglionic sympathetic neurons

Neurotransmitter Distribution

- Preganglionic neurons (both divisions) release ACh at ganglia
- Postganglionic parasympathetic neurons release ACh at targets
- Postganglionic sympathetic neurons release NE at most targets
- Adrenal medulla releases epinephrine (E) and NE into blood as hormones
- This pattern determines the functional effects on target organs



Catecholamine Synthesis

- **Catecholamines** (儿茶酚胺) derived from amino acid **tyrosine** (酪氨酸)
- Biosynthetic pathway:
 - Tyrosine → DOPA → Dopamine → Norepinephrine → Epinephrine
- Each step catalyzed by specific enzymes
- Norepinephrine: Neurotransmitter in sympathetic neurons
- Epinephrine: Hormone from adrenal medulla (has additional methyl group)



Synapses En Passant

- Autonomic axons have **varicosities** (膨体): swellings containing neurotransmitter vesicles
- Form **synapses en passant** (跨越式突触): release occurs along axon length
- Not just at axon terminals
- Allows widespread neurotransmitter release to smooth muscle cells
- Sympathetic and parasympathetic axons often innervate same target cells



Antagonistic Effects on Target Cells

- Same target cells receive both sympathetic and parasympathetic innervation
- Sympathetic axons release norepinephrine → binds **adrenergic receptors** (肾上腺素能受体)
- Parasympathetic axons release acetylcholine → binds **cholinergic receptors** (胆碱能受体)
- Usually produce opposite (antagonistic) effects
- Allows precise regulation through balance of inputs

Adrenergic Receptors

- Two major classes of **adrenergic receptors**:
 - **Alpha (α) receptors**: α_1 and α_2 subtypes
 - **Beta (β) receptors**: β_1 and β_2 subtypes
- All are **G-protein-coupled receptors** (G 蛋白偶联受体)
- Different tissues express different receptor subtypes
- Allows epinephrine and norepinephrine to have diverse effects

Adrenergic Receptor Effects

- α_1 : Vasoconstriction, pupil dilation (radial muscle contraction)
- α_2 : Negative feedback (inhibits NE release)
- β_1 : Increases heart rate and contractility
- β_2 : Bronchodilation, vasodilation in skeletal muscle, uterine relaxation
- Receptor distribution determines tissue-specific responses



Autonomic Receptor Overview

- All preganglionic neurons release ACh → activates **nicotinic receptors** on postganglionic neurons
- Postganglionic parasympathetic neurons release ACh → activates **muscarinic receptors** on targets
- Postganglionic sympathetic neurons release NE → activates **adrenergic receptors** on targets
- Receptor type determines whether effect is excitatory or inhibitory



Clinical: Agonists and Antagonists

- **Agonist** (激动剂): Drug that activates a receptor (mimics neurotransmitter)
- **Antagonist** (拮抗剂): Drug that blocks a receptor (prevents activation)
- Examples:
 - β_2 agonists (albuterol): Treat asthma by dilating bronchioles
 - β -blockers: Reduce heart rate and blood pressure
 - α_1 antagonists: Treat hypertension by causing vasodilation

Cholinergic Receptors

- Two types of **cholinergic receptors**:
 - **Nicotinic receptors** (烟碱型受体): Activated by nicotine
 - **Muscarinic receptors** (毒蕈碱型受体): Activated by muscarine
- Located in different tissues
- Have different structures and mechanisms
- Mediate different physiological effects

Nicotinic vs Muscarinic Receptors

- **Nicotinic receptors:**
 - **Ligand-gated ion channels** (配体门控离子通道)
 - ACh binding directly opens channel
 - Na^+ influx, K^+ efflux $\rightarrow$ depolarization
 - Always excitatory
 - Blocked by curare
- **Muscarinic receptors:**
 - **G-protein-coupled receptors**
 - ACh binding activates G-proteins
 - Indirectly affects ion channels and enzymes



Nicotinic vs Muscarinic Receptors

- Can be excitatory or inhibitory
- Blocked by atropine

Cholinergic Receptor Distribution

- **Nicotinic receptors:**

- Skeletal muscle neuromuscular junction
- All autonomic ganglia (both divisions)
- Always cause depolarization and excitation

- **Muscarinic receptors:**

- All postganglionic parasympathetic target organs
- Five subtypes (M_1 - M_5) with different effects



Cholinergic Receptor Distribution

- M_3/M_5 : Smooth muscle contraction, gland secretion
- M_2 : Heart rate slowing (hyperpolarization)

Clinical: Atropine Effects

- **Atropine** (阿托品) is a muscarinic receptor antagonist
- Blocks parasympathetic effects at target organs
- Clinical effects:
 - Pupil dilation (mydriasis)
 - Increased heart rate
 - Dry mouth (reduced salivation)
 - Reduced GI motility
- Used to treat bradycardia, reduce secretions before surgery

Other Autonomic Neurotransmitters

- Some autonomic neurons are **nonadrenergic, noncholinergic** (NANC)
- Use alternative neurotransmitters:
 - **ATP** (三磷酸腺苷): Purinergic transmission
 - **Vasoactive intestinal peptide** (VIP, 血管活性肠肽)
 - **Nitric oxide** (NO, 一氧化氮)
- Provide additional complexity and specificity to autonomic control

Nitric Oxide as Neurotransmitter

- **Nitric oxide (NO)** is a gaseous neurotransmitter
- Produced by some parasympathetic neurons
- Not stored in vesicles; synthesized on demand
- Ca^{2+} activates **nitric oxide synthetase** → converts L-arginine to NO
- Effects:
 - Penile erection (vasodilation)
 - Cerebral vasodilation
 - GI smooth muscle relaxation
- Also acts as **paracrine regulator** in tissues

Dual Innervation: Overview

- Most organs receive **dual innervation** (双重神经支配)
- Innervated by both sympathetic and parasympathetic divisions
- Three types of interactions:
 - **Antagonistic** (拮抗作用): Opposite effects
 - **Complementary** (互补作用): Similar effects
 - **Cooperative** (协同作用): Different effects working together

Antagonistic Effects

- **Heart:**
 - Sympathetic → increases rate and contractility (β_1)
 - Parasympathetic → decreases rate (M_2 on vagus)
- **Digestive tract:**
 - Sympathetic → inhibits motility and secretion
 - Parasympathetic → promotes motility and secretion
- **Iris:**
 - Sympathetic → pupil dilation (radial muscle)
 - Parasympathetic → pupil constriction (circular muscle)

Complementary Effects

- **Salivary glands:**
 - Parasympathetic → watery saliva production
 - Sympathetic → vasoconstriction in gland
 - Result: Thicker, more viscous saliva from sympathetic
- Both divisions contribute to salivary function
- Produce different qualities of saliva for different situations

Cooperative Effects

- **Male sexual function:**
 - Parasympathetic → penile erection (vasodilation)
 - Sympathetic → ejaculation
- **Female sexual function:**
 - Parasympathetic → clitoral erection, vaginal secretions
 - Sympathetic → orgasm
- **Micturition** (排尿):
 - Parasympathetic → bladder contraction
 - Sympathetic → increased bladder muscle tone
- Both divisions required for complete function

Organs Without Dual Innervation

- Some organs receive only **sympathetic innervation**:
 - Adrenal medulla
 - **Arrector pili muscles** (立毛肌, “goose bumps”)
 - Sweat glands
 - Most blood vessels
- Regulation by varying sympathetic tone (firing rate)
- Increased tone → activation
- Decreased tone → reduced activity or opposite effect

Thermoregulation

- Temperature regulation uses only sympathetic division
- **In heat:**
 - Decreased sympathetic → cutaneous vasodilation → heat loss
 - Increased sweating (via cholinergic sympathetic fibers)
- **During exercise:**
 - Cutaneous vasoconstriction in limbs
 - Increased trunk sweating
- **After exercise:**
 - Decreased sympathetic → vasodilation → heat elimination
- Parasympathetic not involved in thermoregulation

Higher Brain Control

- ANS is regulated by higher brain centers (not truly “autonomous”)
- Key regions:
 - **Medulla oblongata** (延髓): Cardiovascular and respiratory centers
 - **Hypothalamus** (下丘脑): Master autonomic regulator
 - **Limbic system** (边缘系统): Emotional responses
 - **Cerebral cortex** (大脑皮层): Cognitive/emotional influence

Visceral Reflexes

- Sensory information from viscera enters CNS via afferent fibers
- Creates reflex arcs for automatic autonomic responses
- Examples:
 - **Baroreceptor reflex**: Regulates blood pressure
 - Stretch reflexes in hollow organs
 - Chemoreceptor reflexes
- Occur at spinal cord and brainstem levels



Fight-or-Flight Response

- **Sympathetic activation** in response to stress or danger
- Mediated by hypothalamus and limbic system
- Coordinated effects:
 - Increased heart rate and blood pressure
 - Bronchodilation (increased air flow)
 - Pupil dilation (improved vision)
 - Decreased digestion
 - Increased blood glucose (energy mobilization)
 - Adrenal medulla releases epinephrine

Rest-and-Digest Response

- **Parasympathetic activation** during rest and feeding
- Promotes energy conservation and restoration
- Effects:
 - Decreased heart rate
 - Increased digestive activity (motility and secretion)
 - Increased nutrient absorption
 - Energy storage (glycogen, fat synthesis)
 - Pupil constriction
 - Promotes elimination (urination, defecation)

Integration and Homeostasis

- Balance between sympathetic and parasympathetic activity maintains homeostasis
- Shift toward sympathetic: Preparation for activity, stress response
- Shift toward parasympathetic: Recovery, restoration, digestion
- Integration with endocrine system (**neuroendocrine integration**, 神经内分泌整合)
- Hypothalamus coordinates autonomic and hormonal responses
- Essential for survival and adaptation to environment